CS-217 Quiz 3

Date: January 28, 2026
Duration: 20 minutes

Q1 (1 mark). Let $w \in \mathbb{R}^d$ be the parameter vector. Which of the following priors on the parameter vector w leads to L2 regularization under MAP estimation?

1. $w \sim \text{Uniform}(\mathbb{R}^d)$
2. $w \sim \text{Laplace}(0, b)$
3. $w \sim \mathcal{N}(0, \sigma^2 I)$
4. $w \sim \text{Cauchy}(0, \gamma I)$

Solution: 3. $w \sim \mathcal{N}(0, \sigma^2 I)$
Under MAP estimation, using a Gaussian prior $w \sim \mathcal{N}(0, \sigma^2 I)$ gives L2 regularization. Since the log-density of the Gaussian prior is proportional to $-\|w\|_2^2$ giving us the penalty term. The variance becomes the regularization parameter.

Q2 (1 mark). Let $\log f(t)$ be convex for $t \in [0, 1]$. Assume $\alpha_t = f(t)$. Which of the following inequalities hold for α_t?

1. $\dfrac{\alpha_{t+1}}{\alpha_t} \geq \dfrac{\alpha_t}{\alpha_{t-1}}$
2. $\alpha_{t+1} - \alpha_t \geq \alpha_t - \alpha_{t-1}$
3. $\alpha_{t+1} \geq \alpha_t$

Solution: 1. $\dfrac{\alpha_{t+1}}{\alpha_t} \geq \dfrac{\alpha_t}{\alpha_{t-1}}$
Since $\log f(t)$ is convex, it satisfies the convexity inequality

$$\log f\left(\tfrac{t+1+t-1}{2}\right) \ \leq \ \tfrac{1}{2} \log f(t+1) + \tfrac{1}{2} \log f(t-1).$$

This gives

$$\log \alpha_t \leq \tfrac{1}{2}\left(\log \alpha_{t+1} + \log \alpha_{t-1}\right),$$

or equivalently,

$$2 \log \alpha_t \leq \log \alpha_{t+1} + \log \alpha_{t-1}.$$

Exponentiating both sides yields

$$\alpha_t^2 \leq \alpha_{t+1}\alpha_{t-1} \quad \Longleftrightarrow \quad \frac{\alpha_{t+1}}{\alpha_t} \geq \frac{\alpha_t}{\alpha_{t-1}}.$$

Hence, option (1) holds.

Q4 (1 mark). Which of the following methods can suffer from overfitting?

1. Maximum a Posteriori Estimation (MAP)
2. Maximum Likelihood Estimation (MLE)
3. Full Bayesian Estimation

Solution: 2. Maximum Likelihood Estimation (MLE)

Q5 (1 mark). In gradient descent, we take the gradient of the loss function to find the direction of greatest increase in loss. Moving in this direction reduces the loss. The above statement is:

1. True
2. False

Solution: 2. False
Gradient gives the direction of maximum increase thus moving in the opposite direction will reduce the loss. This what we do in SGD.

Q6 (1 mark). Consider the Lagrangian

$$\mathcal{L}(w, u) = F(w) + ug(w), \qquad u \geq 0$$

constrained such that $g(w) \leq 0$. Suppose $F(w)$ is finite for all w. Fix some w_0 such that $g(w_0) > 0$. What can be concluded about

$$\max_{u \geq 0} \mathcal{L}(w_0, u)?$$

1. It equals $F(w_0)$
2. It is finite and larger than $F(w_0)$]
3. It is unbounded above
4. It equals zero

Solution: 3. It is unbounded above
When w_0 is fixed, the function $\mathcal{L}(w_0, u)$ can increase indefinitely as $u \to \infty$.

Q7 (1 mark). Suppose a proximal update produces θ_{t+1} satisfying

$$||\theta_{t+1} - \tilde{\theta}|| \leq ||\theta - \tilde{\theta}||_2 \quad \forall \theta \in S$$

What can be concluded from the above? Select all the possible values.

1. θ_{t+1} minimizes $\mathcal{L}(\theta)$ over S

2. θ_{t+1} is the projection of $\tilde{\theta}$ onto S

3. θ_{t+1} satisfies the KKT conditions for the original problem

4. S must be convex

Solution: 2. θ_{t+1} is the projection of $\tilde{\theta}$ onto S
The equation describing θ_{t+1} justifies that it is the closest to $\tilde{\theta}$ in S.

Q8 (1 mark). let $F : \mathbb{R}^d \to \mathbb{R}$ be convex and minimized at $w = 0$ without constraints. Now consider the constrained problem

$$\min_{w} F(w) \quad : c^T w \leq a, c \in \mathbb{R} \backslash \{0\}$$

For which values of a is it certain that the constraint does NOT affect the optimizer? Select all the possible values.

1. $a = 0$

2. $a < 0$

3. $a > 0$

4. All values of a

Solution: 1,3
Keeping $w = 0$, putting $a \geq 0$ is automatically satisfies the constrained optimization problem.

Q9 (1 mark). Suppose w^* minimizes $F(w)$ subject to a single constraint $g(w) \leq 0$, and the constraint is active at w^*. Assume both gradients exists and $\nabla g(w^*) \neq 0$. Which statement must hold at w^*?

1. $\nabla F(w^*)$ points strictly into the feasible region

2. $\nabla F(w^*)$ is orthogonal to $\nabla g(w^*)$

3. $-\nabla F(w^*)$ is a non-negative scalar multiple of $\nabla g(w^*)$

4. $\nabla g(w^*) = 0$

Solution: 3
Applying KKT condition on the constraint we get $\nabla F(w^*) + ug(w^*) = 0 \quad ; u \geq 0$, implies (3) is correct. Other options are not necessarily true.